



Near-infrared and upconversion luminescence of Tm^{3+} and $\text{Tm}^{3+}/\text{Yb}^{3+}$ -doped oxyfluorosilicate glasses

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ABSTRACT

Oxyfluorosilicate glasses-doped with different concentrations of Tm^{3+} and $\text{Tm}^{3+}/\text{Yb}^{3+}$ ions have been synthesized and characterized their absorption, photoluminescence, luminescence decay and upconversion properties. The Judd-Ofelt intensity parameters have been obtained from the absorption spectrum of Tm^{3+} single-doped glass and are in turn used to calculate radiative properties such as transition probabilities, branching ratios and lifetimes for the fluorescent levels of Tm^{3+} ions. Visible and near-infrared emission spectra of Tm^{3+} and $\text{Tm}^{3+}/\text{Yb}^{3+}$ -doped glasses have been measured under 473 nm laser excitation. The quenching of emission intensities at higher concentrations of Tm^{3+} ions has been found to be due to cross-relaxation mechanisms. The $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped glasses show upconverted emissions under 975 nm laser excitation. The mechanism of upconversion has been explained based on the pump power dependence of upconversion luminescence. The results reveal that the co-doped glasses are promising for optical fiber amplifiers operating at the relatively low-loss wavelength regions.

1. Introduction

In recent years, researchers focused on oxyfluoride glasses due to their combined advantages of both oxide and fluoride matrices having high stability and low phonon energies, respectively [1–3]. Oxyfluoride glasses doped with trivalent rare-earth (RE^{3+}) ions exhibit attractive luminescence properties, with high transparency, and are found to be potential materials for upconversion luminescence and optical amplification [4–8]. Trivalent thulium (Tm^{3+}) is one of the most attractive RE^{3+} ions that gives visible and infrared luminescence in glasses. Particularly, 1.2 μm emission corresponding to the $^1\text{G}_4 \rightarrow ^3\text{H}_4$ transition of Tm^{3+} ions is useful for optical fiber amplifiers [9].

Spectroscopic investigations on Tm^{3+} -doped glass systems that include silicate [1,5,10–12], bismuth silicate [13], phosphorous silicate [14], fluorophosphate [15–17], metaphosphate [18], borate [19], fluoroborate [20,21], lead borate [3], phosphate [22–24], borophosphate [25], fluoroindate [26], tellurite [27–30], bismuthate [31], fluorozirconate [32], gallate-bismuth-lead [33], heavy metals [34] and germinate [25,35–37] have been reported. Because of simple energy level scheme, long lifetime and large absorption and emission cross-

sections, ytterbium (Yb^{3+}) is used as a sensitizer in RE^{3+} -doped glasses, and co-doping of Tm^{3+} with Yb^{3+} significantly enhances the emission intensities of Tm^{3+} ion [9]. Recently, $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped glass systems that include phosphate [38], fluorophosphate [9,39], fluoride, gallate, aluminate, germinate [39], silicate [39,40], tellurite [41,42], leadfluoroborate [43] and fluorozirconate [44] have been reported. It is worth noting that there are only a few reports on the 1.2 μm emission in Tm^{3+} and $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped oxyfluorosilicate glasses.

The concentration of hydroxyl (OH^-) impurities to an acceptable level in silica fibers has been optimized to minimize the losses (~ 0.5 dB/km) at around 1.2 μm wavelength region [45]. However, not much attention has been paid for the study of emission/amplification at ~ 1.2 μm wavelength region. Ytterbium fiber laser pumped Raman fiber laser is a powerful method to generate ~ 1.2 μm laser action, but it is not a broad band light source and requires a very long fiber Raman cavity [27,46].

From the energy level diagram, Tm^{3+} ion has got two metastable excited states, $^1\text{D}_2$ and $^1\text{G}_4$, which emits blue emission either from their normal emission or by upconversion processes [3]. The latter processes involves the conversion of near-infrared (NIR) light into visible one

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which attracts much attention, in recent times, in the design of optical devices. In optical telecommunications, the RE³⁺-doped fiber amplifiers are playing a vital role [15,41]. Silicate hosts are very interesting because it is easier to integrate into existing silica fiber networks with low transmission loss. Further, silicate glasses are chemically and mechanically stable and can be fabricated from bulk (rod) to a miniature (fiber) size [6,11,47]. In addition, oxyfluoride glasses possess combined advantages of oxide as well as fluoride glasses like low phonon energy, excellent chemical durability and high mechanical strength. Recently, Zhou et al. [27,46] demonstrated the optical amplification at around 1.2 μm based on Tm³⁺: ¹G₄ → ³H₄ emission in tellurite and gallogermanate glasses.

The present work reports the absorption and emission spectra, as well as lifetimes of the emitting levels, of Tm³⁺ ions in single and Tm³⁺/Yb³⁺ co-doped oxyfluorosilicate glasses, including the fluorescence characteristics of 1.2 and 1.6 μm emissions. Upconversion luminescence properties of Tm³⁺/Yb³⁺ co-doped oxyfluorosilicate glasses with 0.1 mol% of Tm³⁺ and 0.1, 0.5, 1.0 and 2.0 mol% of Yb³⁺ have also been studied.

2. Experimental details

Oxyfluorosilicate glasses with the composition (in mol%) of 41 SiO₂ + 10 Al₂O₃ + (26-x) LiF + 23 SrF₂ + x Tm₂O₃ (x = 0.01, 0.05, 0.1, 0.5 and 1.0, denoted as SALSTmx) and 41 SiO₂ + 10 Al₂O₃ + (25.9-x) LiF + 23 SrF₂ + 0.1 Tm₂O₃ + x Yb₂O₃ (x = 0.1, 0.5, 1.0 and 2.0, denoted as SALSTm0.1Ybx) were prepared by the conventional melt quenching technique. Batches of 25 g chemicals were well-mixed in an agate mortar and melted in platinum rhodium crucible with lid at 1450 °C (1723 K) for 2 h. Then the glass melt was poured on to a preheated brass mold and annealed at 450 °C (723 K) for 18 h and then cooled down to room temperature to remove thermal stress and strain. After cutting, grinding and polishing, these glass samples were used for optical characterization. Densities of the prepared glasses were measured by the Archimedes method using distilled water as an immersion liquid. Refractive indices were measured using an Abbe-refractometer using 1-bromonaphthalene as contact liquid. Absorption spectra in UV-visible-NIR region were measured using a spectrophotometer (Cary 5000). Visible and NIR emission spectra were obtained by exciting at 473 nm using diode pumped solid-state laser and analyzing the emission with a 500 mm focal length spectrograph (Andor Shamrock 500i) equipped with intensified CCDs and InGaAs cameras. Decay curves for the ¹G₄ and ³H₄ levels were measured by exciting at 473 and 776 nm laser, respectively, using optical parametric oscillator (EKSPLA/NT342/3/UVF) and the signal was detected with the help of a digital oscilloscope (Tektronix 2430). All measurements were done at room temperature and were corrected from instrument responses.

For computation of errors, all the above experiments are measured three times and are used to compute mean and standard deviation (σ) using the expression,

$$\sigma = \sqrt{\frac{\sum (x - y)^2}{N}} \quad (1)$$

where x is the actual measurement reading, y is the mean reading and N is the number of measurements.

3. Results and discussion

3.1. Optical absorption spectra and Judd-Ofelt theory

The absorption spectra of SALSTm1.0 and SALSTm0.1Yb0.1 glasses, shown in Fig. 1, were measured in the wavelength range of 340–2000 nm. As can be seen, the absorption peaks at 356, 465, 658, 683, 790, 1210 and 1643 nm correspond to the transitions from ³H₆

ground level of Tm³⁺ to the ¹D₂, ¹G₄, ³F₂, ³F₃, ³H₄, ³H₅ and ³F₄ excited levels, respectively. The absorption peak at 1210 nm corresponds to the ³H₆ → ³H₅ transition shows both electric and magnetic-dipole contributions, whereas the other transitions show the electric-dipole contribution only. The ³H₆ → ³F₄ transition is said to be hypersensitive in nature [3], as it is very sensitive to changes in the local environment of the optically active ion, i.e. it obeys the selection rules: |ΔS| = 0, |ΔL| ≤ 2, and |ΔJ| ≤ 2. In addition, the intense band at 976 nm, observed in Tm³⁺/Yb³⁺ co-doped glass, belongs to the absorption of Yb³⁺ ions from the ²F_{7/2} ground state to the ²F_{5/2} excited state. The Judd-Ofelt (JO) theory [48,49] has been widely used to predict the spectroscopic properties of RE³⁺ ions. The JO intensity parameters have been obtained from the absorption spectrum of 1.0 mol% Tm₂O₃-doped glass. The experimental oscillator strengths, *f*_{exp}, of all the absorption bands of Tm³⁺ are determined using the Smakula formula [50]. These *f*_{exp} values are in turn used to evaluate the JO intensity parameters (Ω₂, Ω₄, Ω₆) by a least-square fitting method [20]. The expressions used for the calculation of JO parameters and other radiative properties are given below [18,20,48,49].

The oscillator strength of an absorption transition (*f*_{exp}) is directly proportional to the area under the absorption curve and is expressed as

$$f_{exp} = 4.32 \times 10^{-9} \int \varepsilon(\nu) d\nu \quad (2)$$

where ε(ν) is the molar absorptivity of a band at a wave number ν (cm⁻¹).

According to JO theory, the calculated oscillator strength for the absorption band corresponding to the electronic transition from the ground state, ΨJ, (assumed that all the crystal-field levels of the ground state are equally populated) to an excited state, Ψ'J', is given by

$$f_{cal} = \frac{8\pi^2 m c \nu}{3h(2J+1)} \frac{(n^2+2)^2}{9n} \sum_{\lambda=2,4,6} \Omega_{\lambda} (\Psi J \| U^{\lambda} \| \Psi' J')^2 \quad (3)$$

where 'm' is the mass of the electron, 'c' is the speed of light in vacuum, 'ν' is the wavenumber (cm⁻¹) associated with the transition, 'h' is the Planck's constant, 'n' is the refractive index of the host lattice, $\frac{(n^2+2)^2}{9n}$ is the local field correction for the effective field induced by the host lattice, 'J' is the total angular momentum of the ground state, Ω₂, Ω₄ and Ω₆ are called JO intensity parameters and ||U^(λ)||² are the squared doubly reduced matrix elements evaluated from the intermediate coupling approximation for a transition, ΨJ → Ψ'J'. The oscillator strengths of the various observed transitions are evaluated through Eq. (2) and are used in Eq. (2). A least-square fitting approximation is then used for Eq. (3) to determine Ω_λ (λ = 2, 4 and 6) parameters, which give the best fit between experimental and calculated oscillator strengths. The theoretical oscillator strengths (*f*_{cal}) are then determined using Ω_λ and Eq. (2).

The radiative transition probability (A) for a transition ΨJ → Ψ'J' can be expressed by

$$A(\Psi J, \Psi' J') = \frac{64\pi^4 \nu^3}{3h(2J+1)} \frac{n(n^2+2)^2}{9} S_{ed} + \frac{64\pi^4 \nu^3}{3h(2J+1)} n^3 S_{md} \quad (4)$$

where, S_{ed} and S_{md} represents the electric-dipole and magnetic-dipole line strengths, respectively between the ground state and particular excited state. For electric-dipole transitions,

$$S_{ed} = e^2 \sum_{\lambda=2,4,6} \Omega_{\lambda} (\Psi J \| U^{\lambda} \| \Psi' J')^2 \quad (5)$$

For magnetic-dipole transitions,

$$S_{md} = \frac{e^2 h^2}{16\pi^2 m^2 c^2} (\Psi J \| L + 2S \| \Psi' J')^2 \quad (6)$$

The total radiative transition probability (A_T) for an excited state is the sum of the A(ΨJ, Ψ'J') terms calculated over all the terminal states

$$A_T(\Psi J) = \sum A(\Psi J, \Psi' J') \quad (7)$$

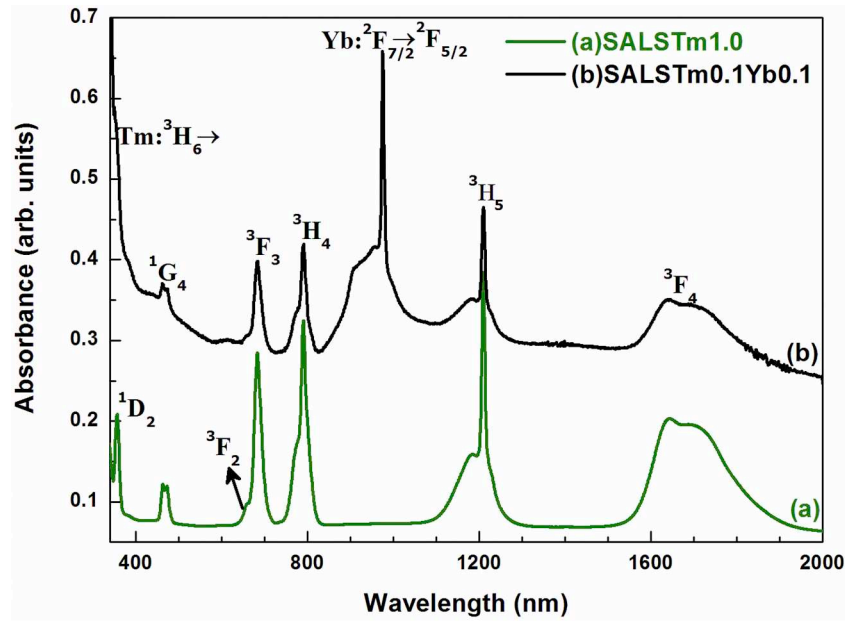


Fig. 1. Absorption spectra of (i) SALSTm1.0 and (ii) SALSTm0.1Yb0.1 glasses in UV- visible-NIR regions.

A_T is related to the radiative lifetime (τ_R) of an excited state by

$$\tau_{rad}(\Psi J) = \frac{1}{A_T(\Psi J)} \quad (8)$$

The branching ratio (β_R) corresponding to the emission from an excited $\Psi'J'$ level to its lower level ΨJ is given by

$$\beta_R(\Psi J, \Psi' J') = \frac{A(\Psi J, \Psi' J')}{A_T(\Psi J)} \quad (9)$$

The peak stimulated emission cross-section, $\sigma(\lambda_p)$ ($\Psi J, \Psi' J'$), between the states ΨJ and $\Psi' J'$ having a probability of $A(\Psi J, \Psi' J')$ can be expressed as

$$\sigma(\lambda_p)(\Psi J, \Psi' J') = \frac{\lambda_p^4}{8\pi c n^2 \Delta\lambda_{eff}} A(\Psi J, \Psi' J') \quad (10)$$

where λ_p is the transition peak wavelength and $\Delta\lambda_{eff}$ is its effective linewidth found by dividing the area of the emission band by its average height.

The Ω_λ values thus obtained are used to calculate the theoretical oscillator strengths f_{cal} of the absorption transitions of Tm^{3+} ion. The f_{exp} and f_{cal} values of absorption transitions in SALSTm1.0 glass along with reported f_{exp} values for SBZT10 [14], NBWT10 [15], PKBAT10 [18], PBAWTm [19], BZL [20] and AT10 [47] glasses are presented in Table 1. Smaller root-mean-square deviation, ($\pm 0.51 \times 10^{-6}$), for the

SALSTm1.0 glass indicates the good fit between experimental and calculated oscillator strengths. The evaluated JO parameters ($\times 10^{-20} \text{ cm}^2$) are $\Omega_2 = 5.55$, $\Omega_4 = 2.95$ and $\Omega_6 = 2.20$. The larger value of Ω_2 suggests more asymmetry at the Tm^{3+} ion site and the larger Tm-O covalence in the SALSTm1.0 glass. The lower value of Ω_6 indicates the higher rigidity of SALSTm1.0 glass matrix. The JO intensity parameters and their trends in different Tm^{3+} -doped glasses that include TS3 [1], STE [2], SPL and SPS [10], NBWT10 [15], PBAWTm [19], TPBL [27], TWTm [28], ZBLANTm [32], SLNS [39] and GeBL [46], are compared in Table 2. As can be seen, the JO parameters of the present glass systems are comparable to those of reported Tm^{3+} :glass systems.

The JO parameters are in turn used to determine radiative properties, such as radiative transition probabilities A , branching ratios β_R , and radiative lifetimes τ_R of the excited states of Tm^{3+} ions in SALSTm glasses under study, and are listed in Table 3. As can be seen, the spontaneous emission probability of the $1.2 \mu\text{m}$ emission ($Tm^{3+}:^1G_4 \rightarrow ^3H_4$) is high (206 s^{-1}), indicating its largest intensity.

The τ_R values for various excited emitting levels (1D_2 , 1G_4 , 3H_4 , and 3F_4) in SALSTm1.0 glass is presented in Table 4, along with those of other reported glass systems. As can be seen, τ_R value for the 1G_4 level in present glass is close to PKBAT [18], higher than that of NBWT10 [15], PBAWTm [19], BZL [20], TWTm [28], TNZLTm [29] and PBATm [38], and smaller than ZBLANTm [32] and SCBLTm [40] glass systems.

Table 1

Transition energies (λ , nm), experimental (f_{exp}) and calculated (f_{cal}) oscillator strengths ($\pm 5\%$, $\times 10^{-6}$) for SALSTm1.0 glass and f_{exp} for some reported Tm^{3+} -doped glasses.

Transition	$\lambda(\text{nm})$	SALSTm1.0 [Present work]		f_{exp}						
		f_{exp}	f_{cal}	ZnTm [14]	NBWT10 [15]	PKBAT [18]	PBAWTm [19]	BZL [20]	AT10 [47]	
$^3H_6 \rightarrow$	1D_2	356	3.34	0.41	–	2.13	–	1.91	–	
	1G_4	465	1.67	0.49	0.84	0.96	0.52	0.70	0.99	
	3F_2	658	0.52	0.13	–	0.20	–	1.48	–	
	3F_3	683	4.34	0.96	2.44 ^a	1.56	4.32 ^a	7.88	7.43 ^a	
	3H_4	790	4.00	1.52	2.78	2.28	3.13	7.09	8.22	
	3H_5	1210	3.51	0.45	1.65	2.00	2.09	4.50	3.91	
	3F_4	1643	4.09	1.21	2.28	3.03	2.24	5.62	3.68	

^a The sum of oscillator strengths for $^3F_{2,3}$ transitions.

Table 2

Comparison of Judd-Ofelt intensity parameters (Ω_2 , Ω_4 and $\Omega_6 \pm 5\%$, $\times 10^{-20} \text{ cm}^2$) and their trends in Tm^{3+} -doped glasses.

Glasses	$\times 10\Omega_2$	Ω_4	Ω_6	Trend
SALSTm1.0 [Present work]	5.57	2.94	2.21	$\Omega_2 > \Omega_4 > \Omega_6$
TS3 [1]	3.08	0.99	0.40	$\Omega_2 > \Omega_4 > \Omega_6$
STE [2]	3.25	1.22	0.38	$\Omega_2 > \Omega_4 > \Omega_6$
SPS [10]	3.60	0.79	0.78	$\Omega_2 > \Omega_4 > \Omega_6$
SPL [10]	3.03	0.92	0.79	$\Omega_2 > \Omega_4 > \Omega_6$
NBWT10 [15]	5.28	2.32	1.16	$\Omega_2 > \Omega_4 > \Omega_6$
PBAWTm [19]	2.93	0.58	1.60	$\Omega_2 > \Omega_6 > \Omega_4$
TPBL [27]	4.72	1.35	1.21	$\Omega_2 > \Omega_4 > \Omega_6$
TWTm [28]	6.8	2.0	2.2	$\Omega_2 > \Omega_6 > \Omega_4$
ZBLANTm [32]	2.57	1.90	0.84	$\Omega_2 > \Omega_4 > \Omega_6$
SLNS [39]	3.26	1.20	0.46	$\Omega_2 > \Omega_4 > \Omega_6$
GeBL [46]	4.65	1.51	1.42	$\Omega_2 > \Omega_4 > \Omega_6$

Table 3

Emission transitions, energies ($\nu \pm 2, \text{ cm}^{-1}$), predicted radiative transition probabilities ($A \pm 2, \text{ s}^{-1}$), branching ratios ($\beta_R \pm 2\%$) and lifetimes ($\tau_R, \mu\text{s}$) for the excited levels of SALSTm1.0 glass.

Transition	ν	A	β_R	τ_R
$^1\text{D}_2 \rightarrow$	$^1\text{G}_4$ 6689	159	0.0068	42
	$^3\text{F}_2$ 12,859	885	0.0379	
	$^3\text{F}_3$ 13,455	888	0.0380	
	$^3\text{H}_4$ 15,401	1462	0.0626	
	$^3\text{H}_5$ 19,760	125	0.0054	
	$^3\text{F}_4$ 22,183	11,990	0.5137	
	$^3\text{H}_6$ 28,011	7833	0.3356	
$^1\text{G}_4 \rightarrow$	$^3\text{F}_2$ 6170	12	0.0061	501
	$^3\text{F}_3$ 6766	54	0.0272	
	$^3\text{H}_4$ 8712	206	0.1030	
	$^3\text{H}_5$ 13,071	668	0.3347	
	$^3\text{F}_4$ 15,494	164	0.0821	
	$^3\text{H}_6$ 21,322	892	0.4469	
$^3\text{F}_2 \rightarrow$	$^3\text{F}_3$ 596	0	0	726
	$^3\text{H}_4$ 2542	13	0.0093	
	$^3\text{H}_5$ 6901	228	0.1653	
	$^3\text{F}_4$ 9324	505	0.3669	
	$^3\text{H}_6$ 15,152	632	0.4586	
$^3\text{F}_3 \rightarrow$	$^3\text{H}_4$ 1946	4	0.0016	425
	$^3\text{H}_5$ 6305	262	0.1117	
	$^3\text{F}_4$ 8728	88	0.0377	
	$^3\text{H}_6$ 14,556	1994	0.8489	
$^3\text{H}_4 \rightarrow$	$^3\text{H}_5$ 4359	23	0.0207	885
	$^3\text{F}_4$ 6782	96	0.0850	
	$^3\text{H}_6$ 12,610	1010	0.8943	
$^3\text{H}_5 \rightarrow$	$^3\text{F}_4$ 2423	6	0.0278	4356
	$^3\text{H}_6$ 8251	223	0.9722	
$^3\text{F}_4 \rightarrow$	$^3\text{H}_6$ 5828	208	1.0000	4814

Table 4

Predicted radiative lifetimes ($\tau_R, \mu\text{s}$) for various transitions ($^1\text{D}_2, ^1\text{G}_4, ^3\text{H}_4$ and $^3\text{F}_4$) in SALSTm1.0 glass along with those of reported Tm^{3+} -doped glasses.

Glass	$^1\text{D}_2$	$^1\text{G}_4$	$^3\text{H}_4$	$^3\text{F}_4$
SALSTm1.0 [present work]	42	501	885	4814
NBWT10 [15]	34	454	642	4028
PKBAT [18]	36	515	932	4168
PBAWTm [19]	31	293	497	3636
BZL [20]	20	200	351	2078
TWTm [28]	–	144	966	185
TZNLTM [29]	–	164	316	1727
ZBLANTm [32]	56	774	1310	6020
PBATm [38]	29	430	882	906
SCBLTM [40]	–	570	980	4990

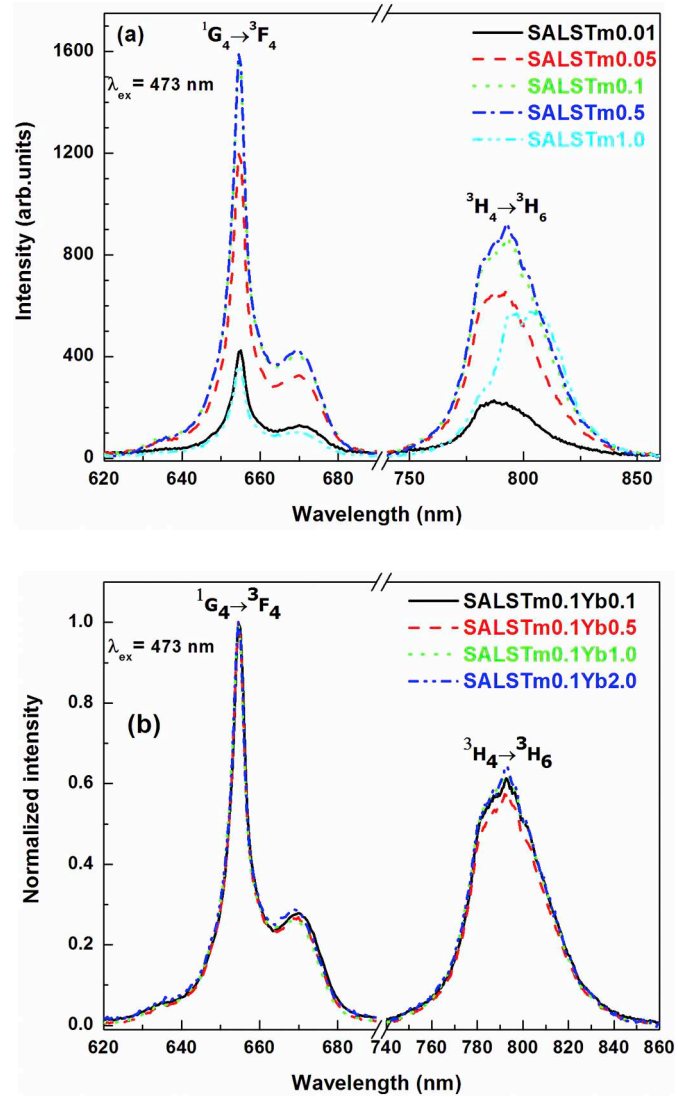


Fig. 2. Visible emission spectra of (a) Tm^{3+} -doped and (b) $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped SALS glasses under 473 nm laser excitation.

Higher τ_R value of a given level indicates its lower spontaneous emission probability and higher stimulated emission probability under suitable conditions. Hence, higher τ_R value of $^1\text{G}_4$ level in SALSTm1.0 glass compared to those of most of the compared systems indicates its higher stimulated emission probability.

3.2. Visible and near-infrared spectra

The visible emission spectra, shown in Figs. 2 (a) and (b) in the wavelength range of 600–900 nm, are measured under the 473 nm laser excitation for the Tm^{3+} and $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped SALS glasses with different concentrations of optically active ions. Fig. 3 shows the partial energy level diagram of Tm^{3+} and Yb^{3+} in SALS glass along with excitation, de-excitation and cross-relaxation channels.

Emission spectra consist of two intense bands at around 650 and 790 nm corresponding to the $^1\text{G}_4 \rightarrow ^3\text{F}_4$ and $^3\text{H}_4 \rightarrow ^3\text{H}_6$ transitions, respectively. As can be seen, in Tm^{3+} single doped glasses, with increase in Tm^{3+} ions concentration from 0.01 to 0.5 mol%, intensities of both ~650 and ~790 nm emissions follow the same increasing trend but for 1.0 mol% doped glass, they decrease non-uniformly. The ~650 nm emission decreases below that of 0.01 mol% doped one and ~790 nm

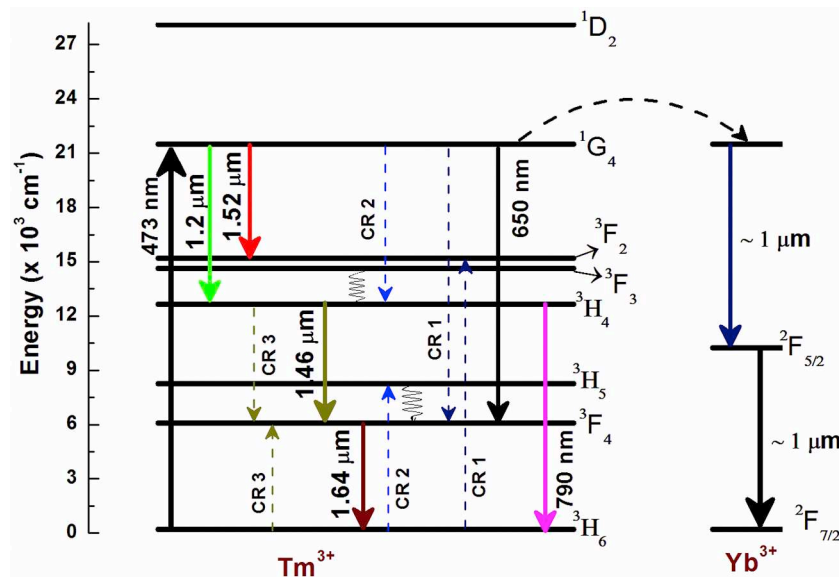


Fig. 3. Partial energy level diagram of Tm^{3+} and Yb^{3+} ions showing transitions of excitation, emission and cross-relaxation channels.

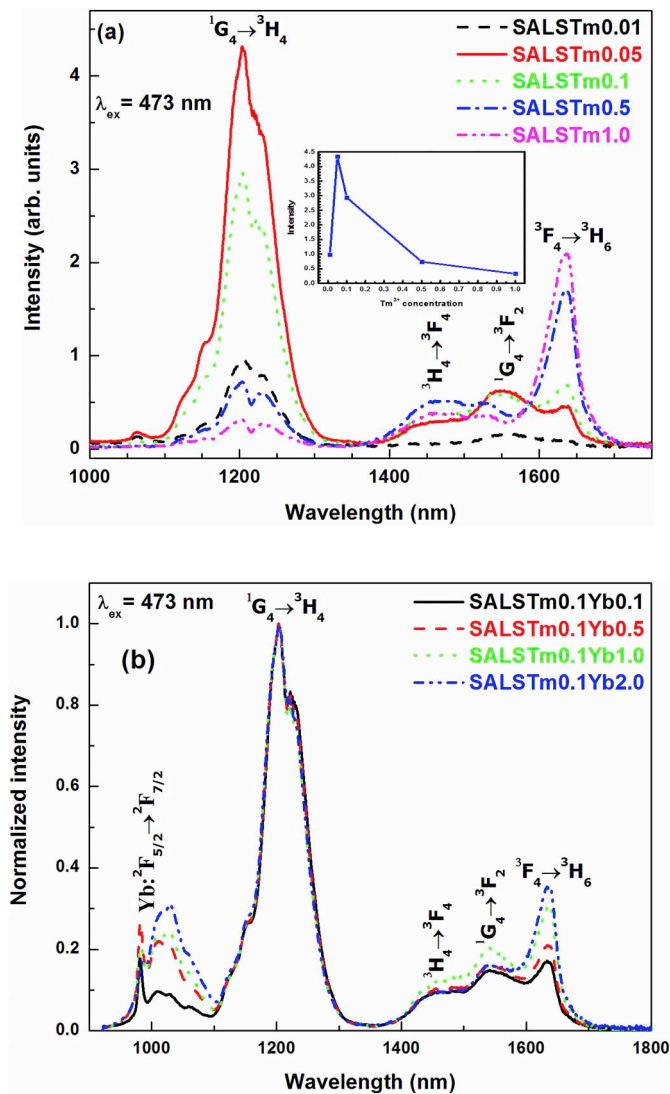


Fig. 4. NIR emission spectra of (a) Tm^{3+} -doped and (b) $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped SALS glasses under 473 nm laser excitation. The inset of the Fig. 4 (a) shows the variation of 1.2 μm emission intensity with Tm^{3+} ions concentration.

emission is > 0.01 mol% doped one but < 0.05 mol% doped glass with slight red shift in band position compared to those of other concentrations. The increase in intensities of the emissions with Tm^{3+} ions concentration up to 0.5 mol% can be explained as due to availability of more ions in the ground state for pumping to excited state. But when the Tm^{3+} ions concentration reaches 1.0 mol%, cross-relaxation energy transfer processes CR1 ($^1\text{G}_4, ^3\text{H}_6 \rightarrow ^3\text{F}_4, ^3\text{F}_2$), CR2 ($^1\text{G}_4, ^3\text{H}_6 \rightarrow ^3\text{H}_4, ^3\text{H}_5$) and CR3 ($^3\text{H}_4, ^3\text{H}_6 \rightarrow ^3\text{F}_4, ^3\text{F}_4$), shown in Fig. 3, becomes dominant resulting in reduced and non-uniform emissions at 650 and 790 nm. The CR1 channel populates the $^3\text{F}_2$ and $^3\text{F}_3$ thermalized levels, which in turn populates the $^3\text{H}_4$ multiplet through multiphonon relaxation and CR2 directly populates $^3\text{H}_4$ level causing increase in the intensity of 790 nm emission. This is also evident from the lowest intensity of the $^1\text{G}_4 \rightarrow ^3\text{H}_4$ transition at 1.0 mol% Tm^{3+} concentration [see Fig. 4(a)], as it will be discussed later. But CR3 process depopulates the $^3\text{H}_4$ level and populates the $^3\text{F}_4$ level causing reduced emission at 790 nm. Energy migration among Tm^{3+} ions and then transfer to traps may also be the reason for overall reduction of emissions of SALSTm1.0 glass. The visible emission spectra of $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped glasses, under 473 nm excitation, is shown in Fig. 2(b) and are normalized to the maximum intensity of the $^1\text{G}_4 \rightarrow ^3\text{F}_4$ transition. As can be seen, the intensity of the $^3\text{H}_4 \rightarrow ^3\text{H}_6$ transition undergoes only marginal changes with increase in Yb^{3+} ions concentration which could be due to small variations in energy transfer processes between Tm^{3+} and Yb^{3+} ions.

The NIR emission spectra of Tm^{3+} and $\text{Tm}^{3+}/\text{Yb}^{3+}$ doped oxy-fluorosilicate glasses with different concentrations of active ions are measured under 473 nm laser excitation and are shown in Figs. 4 (a) and (b), respectively. As can be seen from Fig. 4 (a), there is a broad and intense emission at 1.2 μm associated with the $^1\text{G}_4 \rightarrow ^3\text{H}_4$ transition of the Tm^{3+} ion. The intensity of this emission first increases significantly from 0.01 to 0.05 mol% of Tm_2O_3 and then decreases monotonically up to 1.0 mol% Tm_2O_3 . The initial increase in intensity is due to availability of more number of Tm^{3+} ions for excitation in to the $^1\text{G}_4$ emitting level. The decreasing of intensity of the 1.2 μm emission band at higher Tm^{3+} ion concentrations is attributed to increase in the strength of cross-relaxation CR1 ($^1\text{G}_4, ^3\text{H}_6 \rightarrow ^3\text{F}_4, ^3\text{F}_2$), and CR2 ($^1\text{G}_4, ^3\text{H}_6 \rightarrow ^3\text{H}_4, ^3\text{H}_5$) energy transfer channels (see Fig. 3). The CR energy transfer processes increase with Tm^{3+} ions concentration and lead to the depopulation of the $^1\text{G}_4$ level of Tm^{3+} ions. Zhou et al. also observed similar result of decreasing intensity of the 1.2 μm emission with increase in Tm^{3+} ions concentration in tellurite glasses [27].

In the NIR spectra shown in Fig. 4(a), there are three more

overlapped bands at around 1.46, 1.52 and 1.64 μm regions, assigned to the $^3\text{H}_4 \rightarrow ^3\text{F}_4$, $^1\text{G}_4 \rightarrow ^3\text{F}_2$ and $^3\text{F}_4 \rightarrow ^3\text{H}_6$ transitions, respectively [27]. With increase in Tm^{3+} ions concentration, the 1.64 μm emission increases monotonically from 0.01 to 1.0 mol% which is clear evidence for increase in CR process with increase in Tm^{3+} concentration as all of them are populating the $^3\text{F}_4$ level. The variation of 1.52 μm emission is similar to that of 1.2 μm emission as they are from the same emitting level. The 1.46 μm emission follows the same trend as that of 1.64 μm up to 0.5 mol% of Tm^{3+} ions concentration and then decreases for 1.0 mol% of Tm^{3+} ions. This decrease could be due to activation of the CR3 ($^3\text{H}_4, ^3\text{H}_6 \rightarrow ^3\text{F}_4, ^3\text{F}_4$) channel [39].

To understand the energy transfer process from Tm^{3+} to Yb^{3+} ions in co-doped glasses, emission spectra are measured, exciting at 473 nm, in the range of 925–1750 nm and are shown in Fig. 4(b). The spectra are normalized to the maximum of the 1.2 μm emission. As can be seen, there are emission bands at around 1 μm corresponding to Yb^{3+} ions emission ($^2\text{F}_{5/2} \rightarrow ^2\text{F}_{7/2}$) and their intensities increases continuously with increase in Yb^{3+} ions concentration. The only source of Yb^{3+} ions excitation is energy transfer from the Tm^{3+} ions. As CR1 and CR2 channels populate $^3\text{H}_4$ level of Tm^{3+} ions, the most probable path of energy transfer from Tm^{3+} to Yb^{3+} ions is $\text{Tm} (^3\text{H}_4) + \text{Yb} (^2\text{F}_{7/2}) \rightarrow \text{Tm} (^3\text{H}_6) + \text{Yb} (^2\text{F}_{5/2})$. The amount of energy transfer is found to increase with Yb^{3+} ions concentration, as evidenced from the increase in intensity of the Yb^{3+} ions $^2\text{F}_{5/2} \rightarrow ^2\text{F}_{7/2}$ emission at around 1 μm , which is very much useful for enhancing the efficiency of silicon solar cells [51]. The 1.64 μm emission follows the same trend as the Yb^{3+} ions 1 μm emission, with increase in Yb^{3+} ions concentration, as the energy of the excited $^1\text{G}_4$ level is the source of both the emissions. Intensities of both the 1.46 and 1.52 μm emissions follow that of 1.64 μm emission up to 1.0 mol% and then decreases for 2.0 mol% of Yb^{3+} ions due to increase in strength of CR3 process. Overall, 1.2 μm emission is the dominant one in both the single-doped and co-doped glass systems.

The stimulated emission cross-sections $\sigma(\lambda_p)$ of the intense 1.2 μm emission have been evaluated by using the expression [28],

$$\sigma(\lambda_p) = \frac{\lambda_p^4}{8\pi c n^2 \Delta\lambda_{\text{eff}}} A \quad (11)$$

where λ_p is the peak wavelength; 'n' is the refractive index of the glass; $\Delta\lambda_{\text{eff}}$ is the effective bandwidth; and 'A' is the spontaneous emission probability for the $^1\text{G}_4 \rightarrow ^3\text{H}_4$ transition.

The $\sigma(\lambda_p)$ value for the $^1\text{G}_4 \rightarrow ^3\text{H}_4$ transition in SALSTm0.01, SALSTm0.05 and SALSTm0.1 glasses are presented in Table 5 along with those of reported Tm^{3+} -doped TPBL [27], TWTm [28], GeBL [46] and AT₁₀ [47] glasses. As can be seen, the $\sigma(\lambda_p)$ values of the present glasses decreases with the increase in Tm^{3+} ions concentration and are higher than those of reported ones in TWTm [28], GeBL [46] and AT₁₀ [47] glass systems. For efficient lasers, figure of merit i.e., product of emission cross-section and lifetime ($\text{FOM} = \sigma(\lambda_p) \times \tau_{\text{exp}}$) should be as high as possible. Its value is proportional to the slope efficiency and inversely proportional to the threshold pump power of the laser. The FOM of 1.2 μm emission has been evaluated for three concentrations of Tm^{3+} -doped glasses and are presented in Table 5. As can be seen, FOM

Table 5

Comparison of stimulated emission cross-sections ($\sigma(\lambda_p) \pm 0.2, \times 10^{-21} \text{ cm}^2$) and figure of merit ($\text{FOM}, \times 10^{-24} \text{ cm}^2\text{s}$) for Tm^{3+} : $^1\text{G}_4 \rightarrow ^3\text{H}_4$ transition in Tm^{3+} -doped glasses.

S. No.	Glass	$\sigma(\lambda_p)$	FOM
1	SALSTm0.01 [present work]	3.12	1.43
2	SALSTm0.05 [present work]	3.08	1.16
3	SALSTm0.1 [present work]	3.06	0.85
4	TPBL [27]	4.70	
5	TWTm [28]	5.8	
6	GeBL [46]	2.85	
7	AT ₁₀ [47]	1.2	

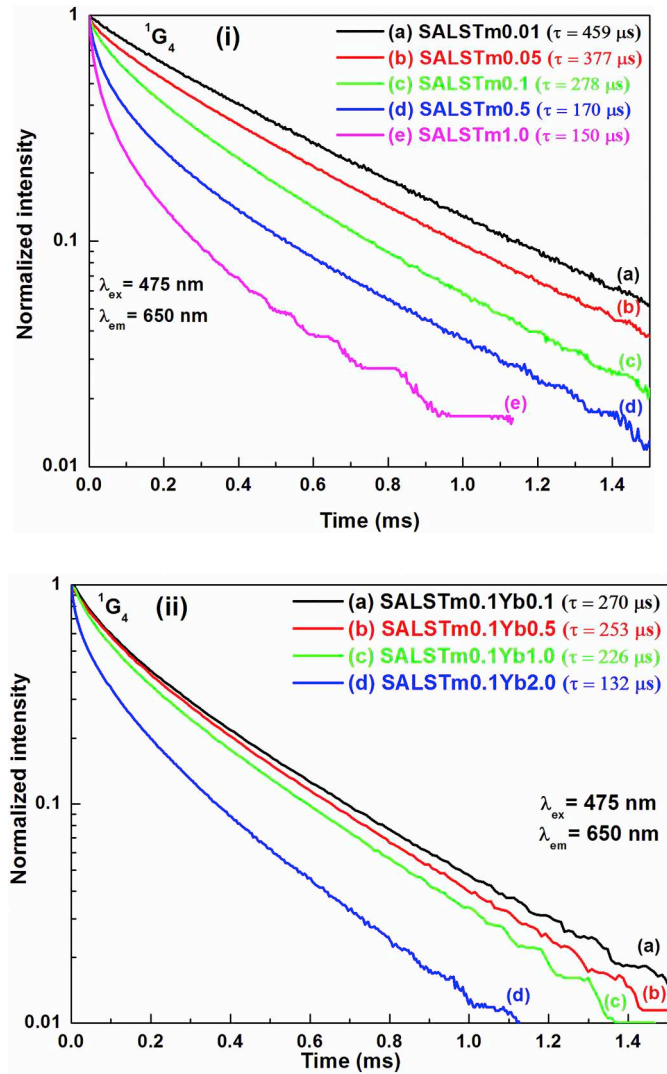


Fig. 5. Decay curves of the $^1\text{G}_4$ level of (i) Tm^{3+} -doped and (ii) $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped SALS glasses under 475 nm laser excitation.

for the 1.2 μm emission is maximum for SALSTm0.01 glass. These results indicate that the present glasses are useful for optical amplification applications at around 1.2 μm wavelength.

3.3. Luminescence decays

Decay curves for the $^1\text{G}_4$ level of Tm^{3+} and $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped glasses under 475 nm laser excitation are shown in Figs. 5(i) and (ii), respectively, whereas Figs. 6(i) and (ii) show the decay curves for the $^3\text{H}_4$ level of Tm^{3+} and $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped glasses, respectively, under 776 nm laser excitation. From Figs. 5 and 6, it is noticed that all the decay curves exhibit non-exponential behavior, indicating the presence of energy transfer processes.

Experimental lifetime (τ_{exp}) values for the $^1\text{G}_4$ and $^3\text{H}_4$ levels in Tm^{3+} -doped glasses and the $^2\text{F}_{5/2}$ level of Yb^{3+} ions in $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped glasses have been evaluated from the measured decay curves using the Eqn. [18],

$$\tau = \int \frac{tI(t)dt}{I(t)} \quad (12)$$

At higher concentrations of Tm^{3+} ions, the interactions between them increase, resulting in a quenching of lifetimes of the $^1\text{G}_4$ and $^3\text{H}_4$ levels. The difference between the values of τ_{exp} , obtained at low concentrations, and τ_R can be understood by the presence of non-radiative

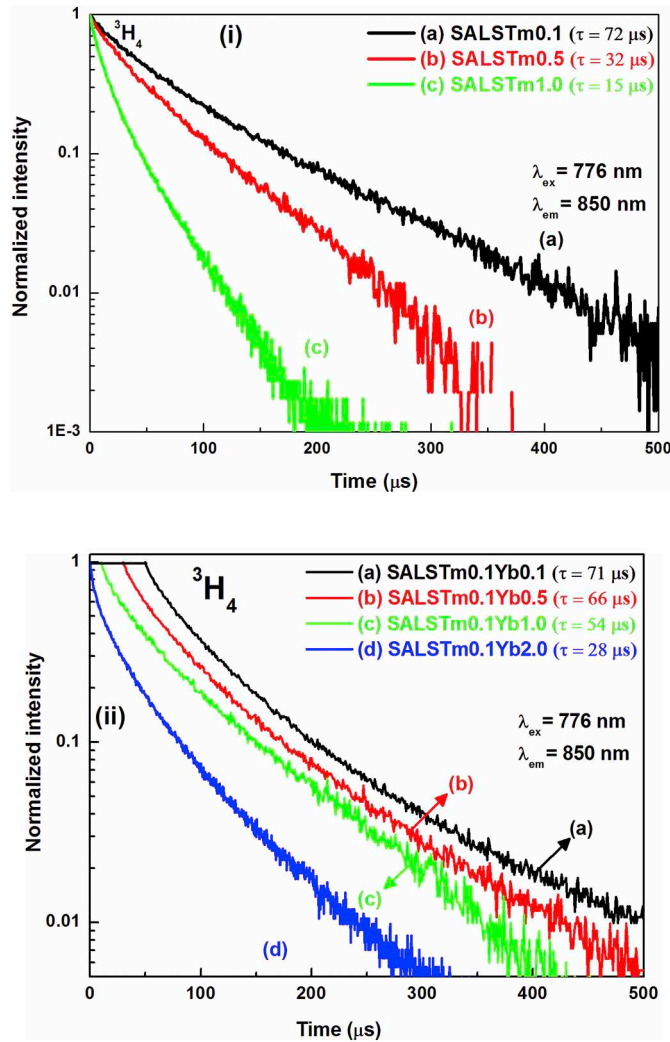
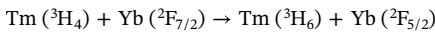
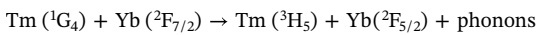


Fig. 6. Decay curves of the 3H_4 level of (i) Tm^{3+} -doped and (ii) Tm^{3+}/Yb^{3+} co-doped SALS glasses under 776 nm laser excitation, monitoring the emission at 850 nm. In Fig (ii) the excitation cutoff has been shifted for clarity.

processes between the transitions [15]. In co-doped samples, lifetime of the $^2F_{5/2}$ level of Yb^{3+} ions decreases with increase in concentration. This could be due to concentration quenching of Yb^{3+} ions due to energy transfer processes between them and then transfer to traps/defects. The decrease in lifetimes of the 1G_4 and 3H_4 levels of Tm^{3+} ions in co-doped systems, even though the concentration of Tm^{3+} ions is kept constant at 0.1 mol%, could be due to the existence of following energy transfer channels from Tm^{3+} to Yb^{3+} ions:



To know the energy transfer from Tm^{3+} to Yb^{3+} ions, the energy transfer efficiency η is an important parameter and it can be calculated using the Eqn. [40],

$$\eta = 1 - \frac{\tau}{\tau_0} \quad (13)$$

where, τ and τ_0 are the experimental lifetimes of the 1G_4 level in co-doped and 0.1 mol% Tm_2O_3 single-doped samples, respectively. The η value for SALSTm0.1Yb2.0 glass is 53% and is higher compared to 32% for zinc barium glass ($0.75Tm^{3+}/2Yb^{3+}$) and 40% for zinc lanthanum glass ($0.75Tm^{3+}/2Yb^{3+}$) [44].

To understand the energy transfer processes effectively, decay

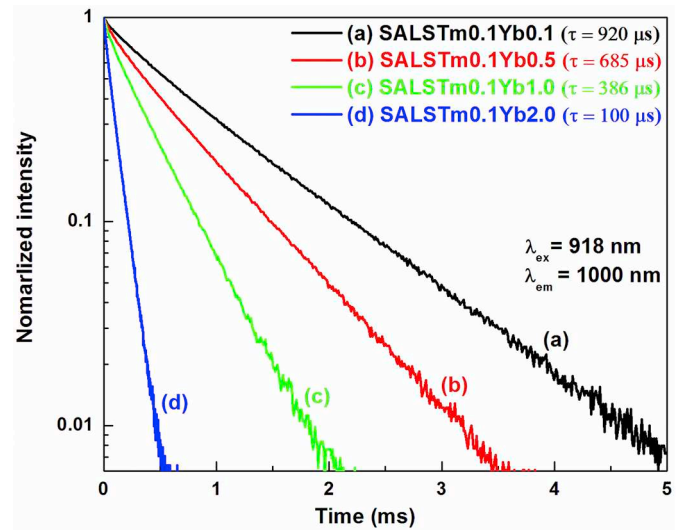


Fig. 7. Decay curves for the $^2F_{5/2}$ level of Yb^{3+} ion in SALSTmYb glasses under 918 nm excitation, monitoring the emission at 1000 nm.

curves for the $^2F_{5/2}$ emitting level of Yb^{3+} ions are measured exciting at 918 nm and monitoring the emission at 1.0 μm in co-doped glasses, and are shown in Fig. 7. The decrease in lifetimes with increase in Yb^{3+} ions concentration is an evidence for the energy transfer from Yb^{3+} to Tm^{3+} ions and/or concentration quenching of Yb^{3+} ions.

3.4. Upconversion luminescence

The upconversion (UC) luminescence is observed in Tm^{3+}/Yb^{3+} co-doped glasses exciting at 975 nm in the wavelength range of 325–700 nm and is shown in Fig. 8. Four emission bands are observed at 362, 450, 474 and 650 nm, which correspond to the $^1D_2 \rightarrow ^3H_6$, $^1D_2 \rightarrow ^3F_4$, $^1G_4 \rightarrow ^3H_6$ and $^1G_4 \rightarrow ^3H_4$ transitions of Tm^{3+} ion, respectively. Among these four emission bands, it is observed that the red emission at 650 nm is relatively weaker than the blue at 474 nm. From Fig. 8, it is observed that the UC luminescence intensity increases with the Yb^{3+} concentration up to 1.0 mol% and decreases thereafter. The variation of 474 and 650 nm UC emission intensities with Yb^{3+} concentration is shown in inset of Fig. 8. For both wavelengths, the UC emission intensity is highest for 1.0 mol% of Yb^{3+} ions and decreases thereafter. The increase in UC luminescence intensity up to 1.0 mol% of Yb^{3+} ions can be explained as due to increase in the efficiency of energy transfer processes from Yb^{3+} to Tm^{3+} ions. The decrease in UC luminescence intensity at higher concentrations of Yb^{3+} ions is caused by concentration quenching through energy migration among the Yb^{3+} ions and then transfer to traps.

In upconversion processes, the relationship between upconversion emission intensity and the intensity of the excitation is given by $I_{UP} \propto I_{IR}^n$, where n is the number of IR photons absorbed per visible photon emitted. The graph between log upconversion intensity (log I_{UP}) and IR excitation intensity (log I_{IR}) shows a straight line with slope n . Fig. 9 shows the plot for log I_{UP} versus log I_{IR} for the 362, 474 and 650 nm emissions. The values of 2.7, 2.0 and 1.9 are obtained for the slopes of 362, 464 and 650 nm emissions, respectively, indicating that emission from the 1D_2 level is a three photon process and emissions from the 1G_4 level is a two photon one [42].

The mechanism of upconversion can be explained using the partial energy level diagram of Tm^{3+} and Yb^{3+} ions shown in Fig. 10. The pump power dependence of UC emission indicates that 1G_4 and 1D_2 levels are populated by absorption of two and three NIR photons, respectively. When the co-doped samples are excited with 975 nm laser, Yb^{3+} ions are excited to the $^2F_{5/2}$ level and energy is transferred to Tm^{3+} ions by three different routes. First one is energy transfer to 3H_5

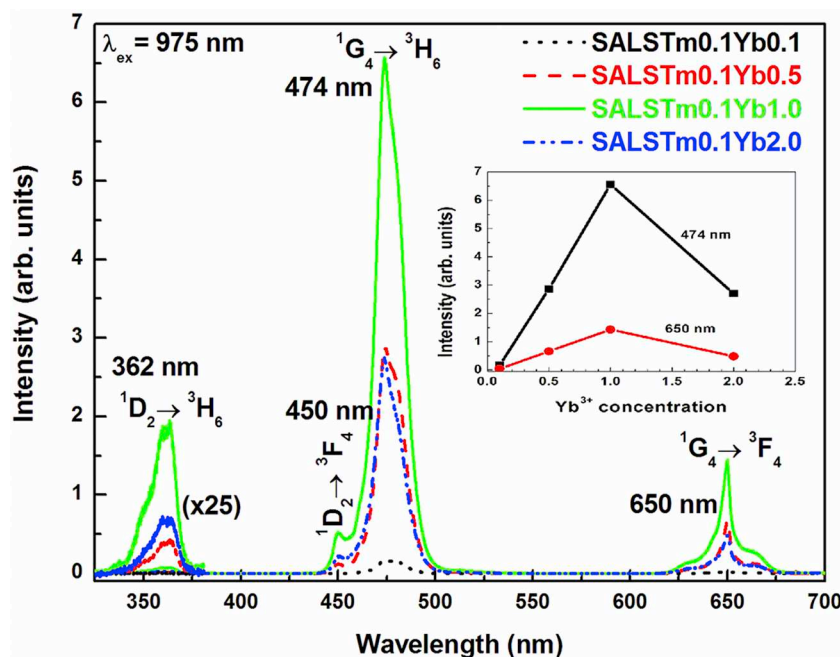


Fig. 8. Upconversion spectra of $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped SALS glasses under 975 nm laser excitation. Inset shows the variation of 474 and 650 nm upconversion emission intensities with Yb^{3+} concentration.

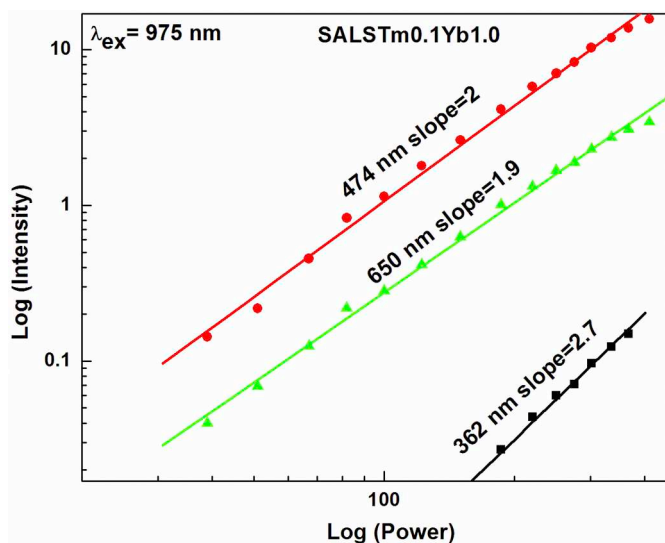


Fig. 9. Pump power dependence of up-converted emissions at 362, 474 and 650 nm in $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped SALS glasses.

level by ET1 process followed by nonradiative relaxation (NR) to the $^3\text{F}_4$ level. Energy transfer upconversion from this level (ETU1) populates the $^3\text{F}_{2,3}$ levels followed by NR to the $^3\text{H}_4$ level. Another possibility is energy transfer from Yb^{3+} ions directly to $^3\text{H}_4$ level of Tm^{3+} through ET2 process and from there another NIR photon populates $^1\text{G}_4$ level (ETU2 or ET3). The $^1\text{G}_4$ level can also be populated by two photon absorption (TPA) at 975 nm from two Yb^{3+} ions [52]. Both processes supports that excitation to the $^1\text{G}_4$ level is a two photon process, reflected in the slope value of 2.0 for the 474 nm emission (Fig. 9). From the $^1\text{G}_4$ level two main radiative decay processes takes place resulting in 650 (red) and 474 nm (blue) emissions corresponding to the $^1\text{G}_4 \rightarrow ^3\text{F}_4$ and $^1\text{G}_4 \rightarrow ^3\text{H}_6$ transitions, respectively.

The ET3 energy transfer from Yb^{3+} ions $^2\text{F}_{5/2}$ level causes ETU3 from $^1\text{G}_4$ to populate the $^1\text{D}_2$ level, although energy matching in this case is poor compared to those of ETU1 and ETU2. Alternatively, the

cross-relaxation (CR) channel, $^1\text{G}_4, ^3\text{H}_4 \rightarrow ^3\text{F}_4, ^1\text{D}_2$, may also populates the $^1\text{D}_2$ level because of a better energy matching. The $^1\text{D}_2$ level also gives two radiative emissions at 362 and 450 nm corresponding to the $^1\text{D}_2 \rightarrow ^3\text{H}_6$ and $^1\text{D}_2 \rightarrow ^3\text{F}_4$ transitions shown in Fig. 8. However, intensities of emissions from the $^1\text{D}_2$ level are relatively weaker than those of the $^1\text{G}_4$ level. This could be due to the weaker efficiency of the excitation processes that populates the $^1\text{D}_2$ level. The three photon excitation process of the $^1\text{D}_2$ level is also reflected in the value of slope (2.7) of the 362 nm emission (Fig. 9).

The dynamics of upconversion has also been investigated by measuring decay curves of the $^1\text{G}_4$ level, exciting at 475 nm and monitoring the emission at 650 nm (Stokes emission); and exciting at 975 nm and monitoring the emission at 475 nm (upconversion), and are shown in Fig. 11 for the SALSTm0.1Yb0.5 glass. As can be seen, in the case of Stokes emission, the lifetime of the $^1\text{G}_4$ level is 253 μs and in the case of upconversion, there is clear rise time of 165 μs and the decay time is 1070 μs , around 4 times more than the normal lifetime. The upconversion decay time reflect population change of feeder level, i.e., $^3\text{F}_4$, which populates $^1\text{G}_4$ through upconversion process. The rise time of upconversion emission reflect the time scale of energy transfer processes [37]. From the above results, it is evident that there is transfer of energy from Yb^{3+} to Tm^{3+} ions under 975 nm excitation because NIR level with longer lifetime ($^3\text{F}_4$) is involved in the upconversion process. On the other hand, if the $^3\text{F}_4$ level is involved in the upconversion, the slopes should be about 3 and 4 for the levels $^1\text{G}_4$ and $^1\text{D}_2$, respectively, but in the present case they are only about 2 and 3. This is because of saturation effect in $\text{Yb}^{3+} \cdot \text{Tm}^{3+}$ upconversion processes which causes population and depopulation of the $^3\text{F}_4$ level [53,54].

4. Conclusions

Oxyfluorosilicate glasses doped with different concentrations of Tm^{3+} and $\text{Tm}^{3+}/\text{Yb}^{3+}$ ions have been prepared and investigated their absorption and emission spectra, lifetimes of the luminescent levels and upconversion properties. From absorption spectrum of the 1.0 mol% Tm_2O_3 -doped glass, intensity analysis has been made using the Judd-Ofelt theory and calculated various radiative properties of the emitting levels and compared with those of similar reported systems. Relative

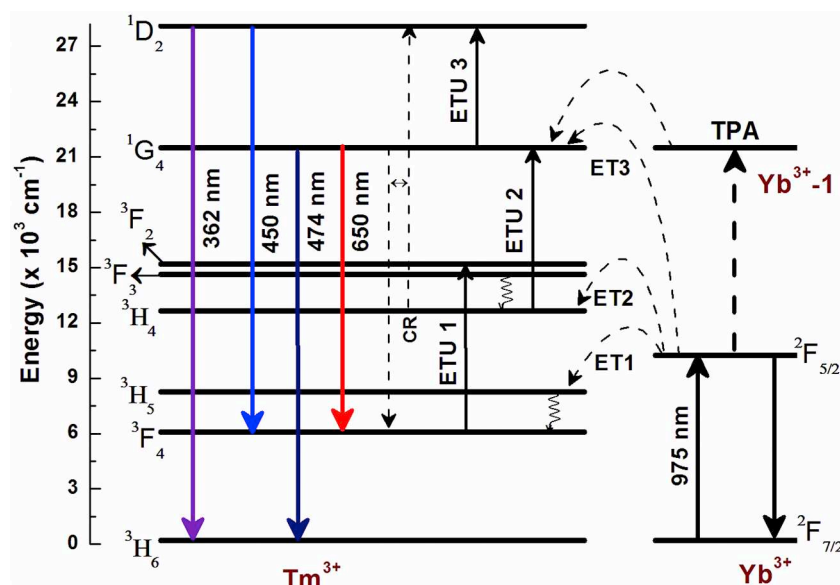


Fig. 10. Partial energy level diagram of Tm^{3+} and Yb^{3+} ions showing the mechanism of upconversion luminescence (TPA: two photon absorption and Yb^{3+-1} represents a second Yb^{3+} ion).

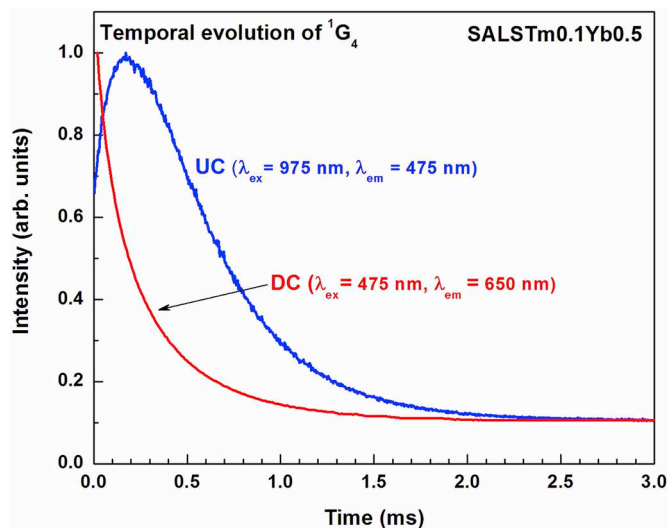


Fig. 11. Temporal evolution of the $^1\text{G}_4$ level emissions in SALSTm0.1Yb0.5 glass under 475 nm excitation (Stokes emission) and under 975 nm excitation (upconversion).

variations of the visible and NIR emission spectra with change in concentrations of active ions have been explained as due to variations of different cross-relaxation processes. Lifetimes of the 650 and 790 nm emissions have been measured and are found to decrease with increase in Tm^{3+} ions concentration in single-doped glasses, due to concentration quenching, and with the increase of Yb^{3+} ions concentration in co-doped glasses due to energy transfer from Tm^{3+} to Yb^{3+} ions. The $1.2 \mu\text{m}$ emission band in NIR spectra of single as well as co-doped glasses is very broad and intense and finds application in optical amplifiers. The NIR emission spectra of co-doped glasses show emission at around $1.0 \mu\text{m}$ which increases with increase in Yb^{3+} ions concentration. The decrease in lifetimes of the $^2\text{F}_{5/2}$ level of Yb^{3+} ions with increase in Yb^{3+} ions concentration is explained as due to energy transfer process/concentration quenching. The $1.0 \mu\text{m}$ emission is useful to enhance Si solar cell efficiency. Finally, the $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped glasses show upconversion emissions under 975 nm excitation. Relative variations of upconversion emissions, their pump power dependencies, mechanism and dynamics have been explained based on energy transfer

processes. Overall, the visible, NIR and upconversion emissions of the studied glasses are very much useful for the design of photonic devices.

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